

Suppressing Tail Emission from $\text{AgIn}_{1-x}\text{Ga}_x\text{S}_2/\text{AgGaS}_2$ Quantum Dots by GaI_3 –Assisted Interface Reinforcement

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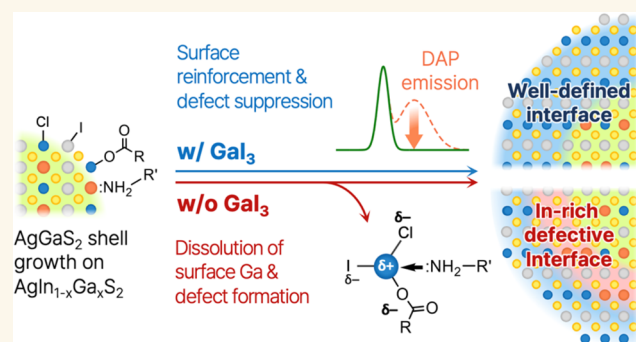
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ABSTRACT: Heteroepitaxy of AgGaS_2 (AGS) or GaS_x shells on $\text{AgIn}_{1-x}\text{Ga}_x\text{S}_2$ (AIGS) quantum dots (QDs) enables narrow band-edge emission with a near-unity photoluminescence quantum yield (QY). While this outcome is attributed to the passivation of donor–acceptor pair states on the surface, persistent tail emission, regardless of the QY, implies underlying interfacial defects in the AIGS/AGS QDs. Herein, we identify the latent compositional and crystallographic disorder at the core–shell interface originating from the surface instability of AIGS QDs. The strong coordination of oleylamine to group III cations promotes partial dissolution of the AIGS cores, resulting in a Ga-deficient, structurally disordered phase that contains interfacial donor states that are thermally mixed with the band-edge electron state. The incorporation of GaI_3 weakens the coordination of oleylamine to cations at the surface and in the dissolution products, thereby suppressing the disorder by fostering a Ga-rich environment. The suppressed tail emission, even at 80 K, highlights the importance of surface stabilization in eliminating the interfacial defects.

KEYWORDS: quantum dot, I–III–VI₂ semiconductor, Ostwald ripening, donor–acceptor pair, low-temperature photoluminescence



Colloidal quantum dots (QDs) have emerged as future display emitters driven by significant advances in colloidal wet chemistry. Central to these advances is the development of core/shell heterostructures.¹ The defect-free heteroepitaxy of wide-band gap (E_g) shells on the QDs yields highly stable and bright emitters with a near unity quantum yield (QY).² The advancement of heterostructured QDs has not only promoted the QD-based display technologies but also expanded their applications to various optoelectronics including solid-state lighting,³ lasers,^{4,5} and single-photon sources.⁶

These efforts are now directed toward environmentally benign QD materials.^{7,8} Recently, I–III–VI₂ compounds-based QDs, such as $\text{AgIn}_{1-x}\text{Ga}_x\text{S}_2$ (AIGS), have emerged as eco-friendly post-InP emitters because of their tunable E_g covering the visible to near-infrared and high absorption cross-section.^{9–12} Although research on heterostructured AIGS QDs is relatively nascent compared to that on orthodox II–VI QDs, the importance of heteroepitaxy remains critical for high photoluminescence (PL) QY. For example, Kuwabata's group demonstrated the passivation of AIGS cores with GaS_x shells using gallium dithiocarbamate in combination with GaCl_3 to enhance gallium reactivity, leading to a high PL QY of 75%

with a narrow PL line width of 31 nm.¹³ Bae's group achieved near-unity PL QY of AIGS QDs by adopting an AgGaS_2 (AGS) shell.¹¹ In this approach, the Ag–S–Ga–(oleate)₂ molecular complex played an important role in the heteroepitaxy of the AGS shell on the AIGS core, preventing the reduction of an Ag precursor during shell growth at high temperatures.

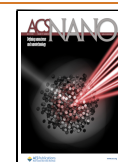
Despite the effective surface passivation of AIGS QDs via modified Ga chemistry, a persistent low-energy tail emission remains observable in their PL spectra, even when the PL QY exceeds 70%.^{11,13,14} This phenomenon implies the presence of defect states capable of radiative recombination even after shell growth. Recent studies on the photophysics of I–III–VI₂ QDs have attributed the origin of the emissive defects to the donor–acceptor pair (DAP) states associated with crystallo-

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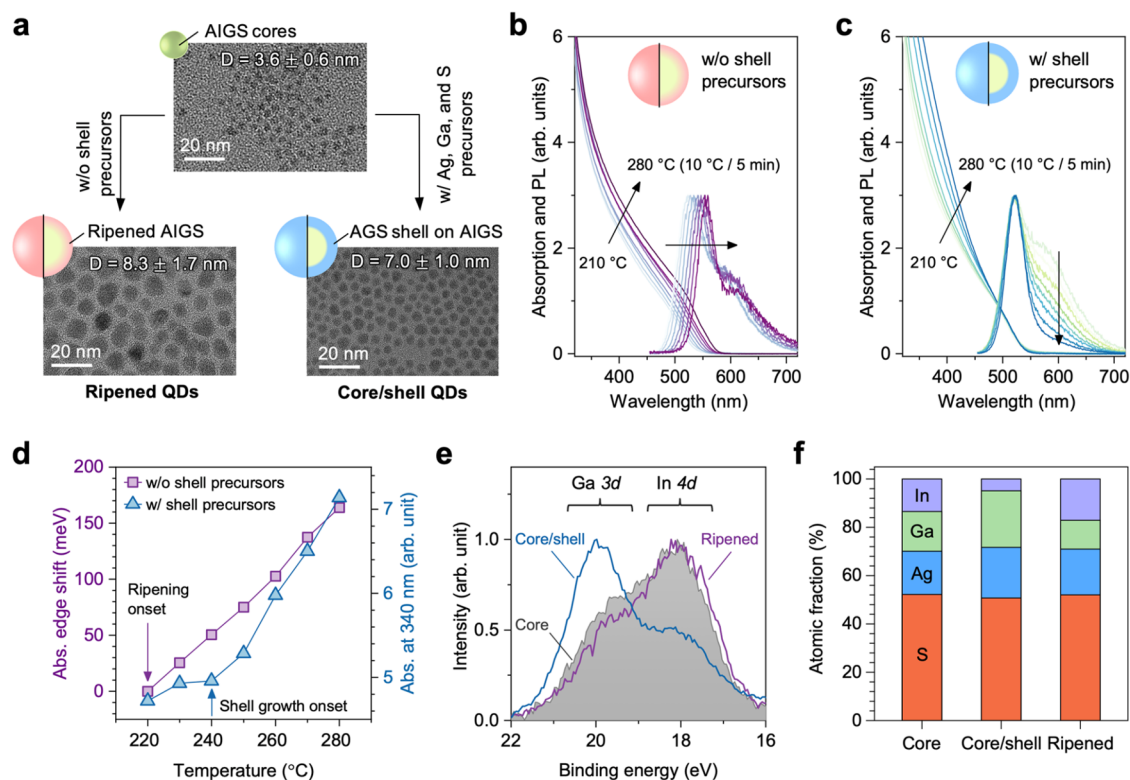


Figure 1. Surface instability of AIGS QDs. (a) TEM images of pristine AIGS cores (top), cores ripened at 280 °C for 30 min without the addition of shell precursors (lower left), and AIGS/AGS core/shell QDs (lower right). (b, c) Evolution of absorption and photoluminescence (PL) spectra with increasing temperature of AIGS QDs (b) without and (c) with the addition of shell precursors. Spectra were recorded at 10 °C intervals. The absorption and PL spectra were normalized to the absorption edge and the band-edge emission, respectively. (d) Absorption edge redshift for ripened AIGS QDs (purple squares, left) absorbance increment at 340 nm for AIGS/AGS QDs during synthesis (blue triangle, right) measured during increasing temperature. Arrows indicate the onset for ripening of AIGS cores (purple arrow) and shell growth reaction (blue arrow). (e) XPS spectra of Ga 3d and In 4d electrons for as-synthesized AIGS (gray background), ripened AIGS cores (purple line), and AIGS/AGS core/shell QDs (blue line). (f) Atomic fractions of Ag, In, Ga, and S in pristine AIGS cores, AIGS/AGS core/shell QDs, and ripened AIGS cores, obtained through EDS.

graphic defects:^{15,16} anion vacancies or antisite defects between group I and III cations are responsible for donor states (ID)), whereas cation vacancies are associated with acceptor states (I A)).¹⁷ Given this interpretation, the narrow band-edge emission of AIGS QDs, recently achieved through surface passivation, suggests that the majority of DAP states are located on the surface of the QDs^{14,18} and can be effectively passivated by Ga.^{19,20} Nonetheless, the persistent tail emission observed even in QDs with near-unity PL QY raises questions regarding (1) the incomplete removal of DAP states and (2) their spatial location within heterostructured QDs (i.e., within the AIGS core, at the AIGS–AGS interface, or on the surface of the AGS shell). In addition to the susceptibility of surface states to the adsorption of ligands such as alkylphosphine,^{13,20} these unresolved issues hamper the understanding and rational heterostructure design of AIGS QDs for better stability, color purity, and efficiency.

Herein, we reveal that the instability of AIGS cores in a coordinating solvent leads to crystallographic disorder at the core–shell interface during the heteroepitaxy reaction, which is responsible for the persistent tail emission frequently observed in AIGS/AGS core/shell QDs. We observed that the AIGS cores are prone to Ostwald ripening in oleylamine (OAm) because of OAm's strong coordination affinity to group III cations. This undesired process disrupts the chemical homogeneity of the ripened QDs owing to the varied

coordination strengths of OAm to different metal cations in the ripening products. Furthermore, ripening induces the generation of an orthorhombic phase. Even in the presence of AGS shell precursors, such a detrimental process results in both compositional and crystallographic disorders, leading to the formation of DAP states at the AIGS–AGS interface and, consequently, a broad tail emission in the PL spectra. Temperature-dependent PL analysis suggests thermal mixing between the band-edge electron state and a high-lying ID that is located approximately 20 meV higher at 300 K. Moreover, the disordered interface in the AIGS/AGS QDs can be mitigated by incorporating an iodide source into the reaction because it suppresses the dissolution of Ga from the QDs by weakening the coordination between OAm and Ga ions. Consequently, the Ga-rich environment on the surface suppresses the formation of DAP states. The AIGS/AGS core–shell QDs synthesized with GaI₃ exhibited significantly reduced tail emission even at 80 K, due to the suppressed involvement of DAP states in the exciton recombination process.

RESULTS AND DISCUSSION

Observation of Surface Instability of AIGS QDs in Coordinating Medium. AIGS QDs were synthesized in coordinating OAm medium using AgI, InCl₃, gallium oleate [Ga(OA)₃] as cation precursors, and sulfur dissolved in OAm

(S-OAm) as anion precursor (details are described in [Methods](#) section). The precursors were mixed at 50 °C and subsequent reaction to 210 °C produced the AIGS cores with a diameter of 3.6 ± 0.6 nm (top image in [Figure 1a](#)). Although achievable PL wavelength of AIGS QDs can be tuned across the full visible range by adjusting the In-to-Ga feed ratio,¹¹ we adopted green-emitting AIGS QDs for this study, as their band-edge emission band is well separated from a broad low-energy tail emission.

While a sufficient amount of ligands is essential for maintaining colloidal stability^{21,22} and regulating the growth kinetics of QDs,^{23,24} their excessive coordination to the surface can possibly promote dissolution or Ostwald ripening^{25–27} that produces monomers or molecular species from QDs.^{28–30} In this context, in the OAm medium the AIGS QDs can be particularly vulnerable, as all constituent elements in AIGS are known to be coordinated with primary amines. To investigate this possibility, we carried out intentional ripening by annealing AIGS QDs in the absence of shell precursors at increasing temperatures. After synthesizing AIGS QDs, we continued to increase the temperature to 280 °C at a rate of 15 °C min^{−1}. At the end of the reaction, the QDs had grown to an average size of 8.3 ± 1.7 nm (lower left image in [Figure 1a](#)), clearly indicating a ripening process that yielded increased and inhomogeneous sizes compared to the initial AIGS QDs. Notably, this increase in size was not due to residual precursors, as demonstrated by a control experiment using purified AIGS cores ([Figure S1](#)). Meanwhile, the AIGS/AGS core/shell QDs synthesized in the presence of shell precursors exhibited an average diameter of 7.0 ± 1.0 nm, corresponding to an AGS shell thickness of approximately 1.7 nm (lower right image in [Figure 1a](#)).

Temporal changes in the absorption and PL spectra of the QDs with and without the addition of shell precursors provide a glimpse of the difference in the activation thresholds of shell growth and ripening. As the temperature increased, the AIGS cores annealed without the addition of shell precursors exhibited red-shifted absorption edges and consistent tailed PL ([Figure 1b](#)). The PL QY of these cores gradually decreased from ~20 to ~6% since 230 °C ([Figure S2](#)). On the other hand, the core/shell QDs retained their absorption edge and suppressed tail emission in the PL spectra, which is a typical signature of AGS shell growth ([Figure 1c](#)). Notably, the AGS shell growth still left a residual tail emission despite a high PL QY of 75%. Meanwhile, the onset temperatures of ripening (based on the absorption edge shift; purple squares in [Figure 1d](#)), and shell growth (based on the absorption enhancement at 340 nm; blue triangles in [Figure 1d](#)) show that ripening begins approximately 40 °C earlier than shell growth, highlighting the thermal instability of AIGS cores in OAm.

Because the ripening process proceeds from pre-existing QDs, the final ripening product is expected to retain the original chemical composition. However, elemental analysis revealed spatial inhomogeneity in the ripened AIGS QDs. To assess this, we used X-ray photoelectron spectroscopy (XPS), which provides compositional information about the entire heterostructure but is surface-sensitive when the QD size is comparable to or larger than the photoelectron escape depth. For the initial AIGS cores and AIGS/AGS core/shell QDs, the Ga 3d and In 4d signals in the XPS spectra closely reflected their bulk compositions because of their small sizes. In contrast, the ripened AIGS QDs with a larger crystalline size (>8 nm), exhibited a slight decrease in the Ga 3d signal and a

broadened In 4d peak, potentially indicating surface enrichment of In ([Figure 1e](#) and [Table S1](#)). To further examine this, we performed elemental analysis using energy-dispersive X-ray spectroscopy (EDS), which provided the elemental ratio of the entire structure ([Figure 1f](#) and [Table S2](#)). Notably, the AIGS QDs displayed an increased In content (from 13.4% to 17.1%) and a decreased Ga content (from 16.4 to 11.9%) after ripening, confirming that the ripened QDs had developed an In-rich surface layer. From this observation, the reduction in PL QY of ripened AIGS QDs can be attributed to the development of an In-rich outer region with a narrower band gap than the core. Such a composition gradient may lead to carrier delocalization toward the surface, thereby enhancing nonradiative exciton recombination. Incorporation of DAP states in this outer region may further promote the likelihood of nonradiative recombination. Other potential causes such as phase segregation into Ag₂S or elemental Ag²⁰ can be excluded based on the uniform distribution of Ag observed in the element mapping of individual ripened QDs ([Figure S3](#)).

The altered chemical composition of ripened QDs introduces important questions about the ripening process: how do ligands participate in the ripening process, and why is the chemical composition altered? Next, we consider the dissolved unidentified species from the QDs as a combination of metal cations, halide anions, and OAm, and relate their affinity to OAm with the chemical content in the ripened QDs.

Halide-Guided Ostwald Ripening. The coordination strength and expected output in the competition of two or more ligands are conveniently described by Pearson's hard and soft acids and bases (HSAB) principle,³¹ which predicts that metal–ligand pairs exhibiting similar hardnesses are preferred. Because the OAm ligand is classified as a hard base,^{32,33} it is expected to coordinate more strongly to Ga³⁺ [chemical hardness (η) = 17] than to In³⁺ (η = 13) or Ag⁺ (η = 6.9).³⁴ This pairing provides insight into the ripening of AIGS QDs and the lowered Ga content observed in the ripened QDs. An intuitive explanation is that the strong coordination of Ga³⁺ with OAm drives the dissolution of QDs into molecules containing Ga³⁺ associated with OAm, and the strong binding between Ga³⁺ and OAm hinders the reincorporation of Ga³⁺ in these molecules into the ripened QDs. However, this simplified interpretation appears to be inconsistent with the synthesis of AIGS QDs: at the beginning of synthesis, Ga³⁺ is already bound to oleate (OA), which is also known as a hard base.³³ Hence, this simplified HSAB concept cannot explain the incorporation of Ga into the intermediate AIS QDs.^{14,35}

To explore the determining factors governing ripening, we first focused on the role of halide ions. Halide-mediated ripening (or sintering) of nanocrystals has been studied in polycrystalline CdTe thin film based on physical vapor deposition^{36–38} and in colloidal nanocrystals.^{39,40} In the solid-state ripening of CdTe grains, chlorine sources form eutectic phases at grain boundaries, and their low melting points promote atomic diffusion and subsequent grain coalescence.⁴¹ In the case of colloidal QDs such as CdS QDs, chloride ions facilitate Ostwald ripening by dissolving surface Cd²⁺ in the form of CdCl₂-tertiary amine complexes, which subsequently contribute to the growth of larger QDs.³⁹ Given these findings, we investigated the influence of halides in our system, particularly those associated with group III elements (In and Ga), because the atomic fractions of S and Ag showed negligible changes after ripening ([Figure 1f](#)).

We supplied four additives (InI_3 , GaI_3 , InCl_3 , and GaCl_3) in the ripening of AIGS QDs, in which the hardness increases from I^- to Cl^- and from In^{3+} to Ga^{3+} . The adsorption of these compounds on the surface can influence the ripening process initiated on the surface and the dissolution-reincorporation of elements in the ripened QDs. As shown in Figure 2, we

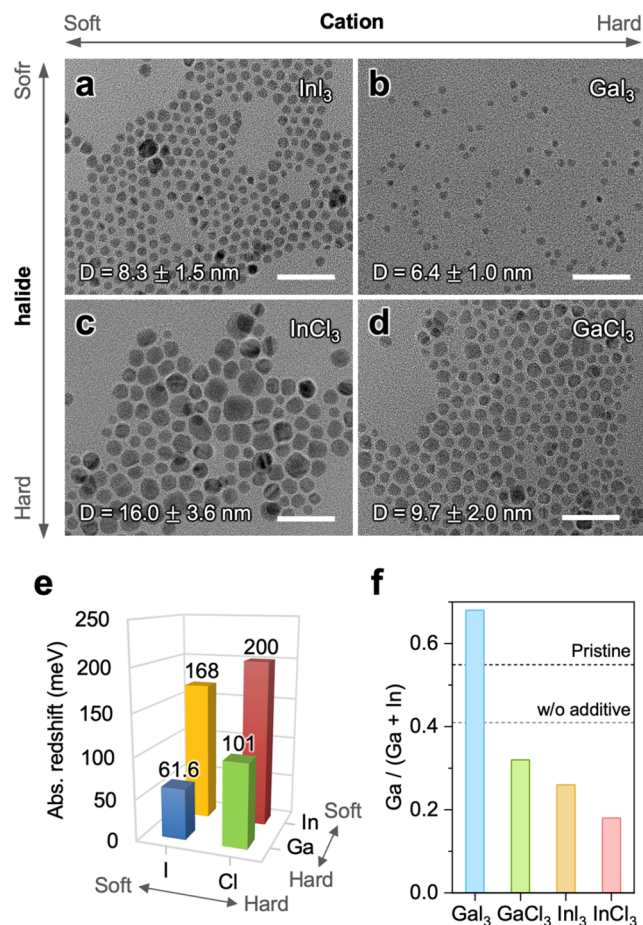


Figure 2. Metal halide-mediated Ostwald ripening of AIGS QDs. (a–d) TEM images of ripened AIGS QDs in the presence of (a) InI_3 , (b) GaI_3 , (c) InCl_3 , and (d) GaCl_3 at 280 °C for 30 min. Scale bar: 50 nm. (e) Absorption edge redshift of ripened QDs with additives in (a–d), referenced to the pristine AIGS cores. (f) Alteration of the Ga-to-In atomic ratio $[\text{Ga}/(\text{Ga} + \text{In})]$ in (a–d), as determined by EDS. Dashed lines indicate the Ga content of pristine (black) and ripened QDs without additive (gray).

assessed ripening by tracking the changes in particle diameter (Figure 2a–d) and the extent of redshift in the absorption edge (Figures 2e and S4). We also examined the variation in Ga content with different additives (Figure 2f). Overall, we confirmed that the GaI_3 , a combination of hard Ga^{3+} and soft I^- , largely suppressed the ripening process as inferred from the lowest optical redshift of 61.6 meV and a marginal size increment from 3.6 to 6.4 nm. Furthermore, this additive only increased the Ga content [in terms of the Ga-to-In atomic ratio $\text{Ga}/(\text{In} + \text{Ga})$] to 68%, which is larger than that of the pristine AIGS cores (55%; see blue bar and black dashed line in Figure 2f). Notably, the Ga content in the ripened QDs could be controlled by the amount of GaI_3 added (Figure S5). On the other hand, InCl_3 amplified the ripening process and Ga extraction, showing the largest size and widest size distribution

(16.0 ± 3.6 nm; Figure 2c), red-shifted absorption edge (200 meV; Figure 2e), and lowest Ga-to-In ratio (18%; Figure 2f). Whereas InI_3 exhibited a comparable size and size distribution with no additives, GaCl_3 exhibited a larger size and wider size distribution (Figure 2a,d). The smaller redshift observed for GaCl_3 compared to InI_3 is attributed to the higher Ga content.

Overall, we conclude that the additives play the following roles in the ripening of AIGS QDs. (i) Halide anions are the main controllers for the extent of ripening, and hard Cl^- intensifies this process. While InI_3 and GaI_3 produced ripened QDs with similar or smaller sizes than those produced without additives, InCl_3 and GaCl_3 resulted in larger sizes. Even after the given reaction period (30 min at 280 °C), the chlorine-mediated ripening continued by dissolving smaller particles. The central role of halides was further confirmed by comparing tetrabutylammonium chloride (TBACl) and tetrabutylammonium iodide (TBAI), which delivered halide anions to the QD surface (Figure S6). We observed that TBACl accelerated the ripening process with an increased size of 11.2 ± 2.2 nm (Figure S6a), whereas the extent of ripening with TBAI was comparable to the cases of InI_3 and no additives (8.2 ± 1.6 nm, Figure S6b). (ii) The addition of cation (In^{3+} or Ga^{3+}) altered the Ga content of the ripened QDs, but its effect was marginal. Except for GaI_3 , all cases displayed a lower Ga content, even when compared to the ripened QDs without additives. (iii) For the same halide, GaX_3 suppressed the extent of ripening. This suppression is attributed to the stability of the lattice during the Ga-exchange process. Because In-to-Ga cation exchange plays a role in the formation of AIGS QDs,³⁵ Ga-deficient conditions make AIGS QDs unstable and susceptible to ripening owing to defect formation during Ga extraction from the lattice.

Understanding the ripening mechanism and role of additives requires an investigation into the molecular structures of the dissolution products. The constituents of the as-synthesized QDs satisfy charge neutrality by pairing cations (i.e., Ag^+ , In^{3+} , and Ga^{3+}), and anions in the lattice (i.e., S^{2-}) and ligands bound on the surface (i.e., halides and oleates). Although OAM stabilizes the metal cations, its neutral L-type character cannot contribute to charge compensation. Assuming that the dissolution products also satisfy charge neutrality in the medium, they are likely combinations of constituent atoms and ligands, hypothetically $[\text{M}(\text{AgS})_a\text{X}_b\text{OAm}_c]_n$ complexes, where $a + b = 3$, $c \geq 0$, $n \geq 1$, $\text{M} = \text{Ga}$ or In , and $\text{X} = \text{Cl}$, I , and/or OA . This product is analogous to the intermediate $\text{Ag-S-Ga}(\text{OA})_2$ observed by Lee et al.¹¹ The inclusion of Ag-S units in the dissolution product is supported by the persistent amounts of Ag and S, regardless of the ripening of the AIGS QDs (Figure 1f and Table S2).

To separate and identify the dissolution products from the residual precursors, the AIGS QDs were purified and dispersed in pure OAM for analysis using matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectroscopy (Figure 3; the assigned peaks are summarized in Table S3). A comparison of the spectra before and after ripening at 280 °C led to two notable observations. First, dispersing the purified QDs in OAM at room temperature even resulted in partial dissolution due to strong coordination affinity of OAM, yielding molecular species such as $\text{Ga}(\text{AgS})_3\text{-OAM}$, $[\text{Ga}(\text{AgS})_2\text{Cl}]_2$, $\text{Ga}(\text{Cl})(\text{I})(\text{OA})\text{-OAM}$, and $\text{Ga}(\text{AgS})\text{Cl}(\text{OA})\text{-OAM}$, as observed in the m/z range of 750–800 (Figure 3). These Ga^{3+} -containing species, coordinated primarily with hard anions, were further intensified after ripening and became

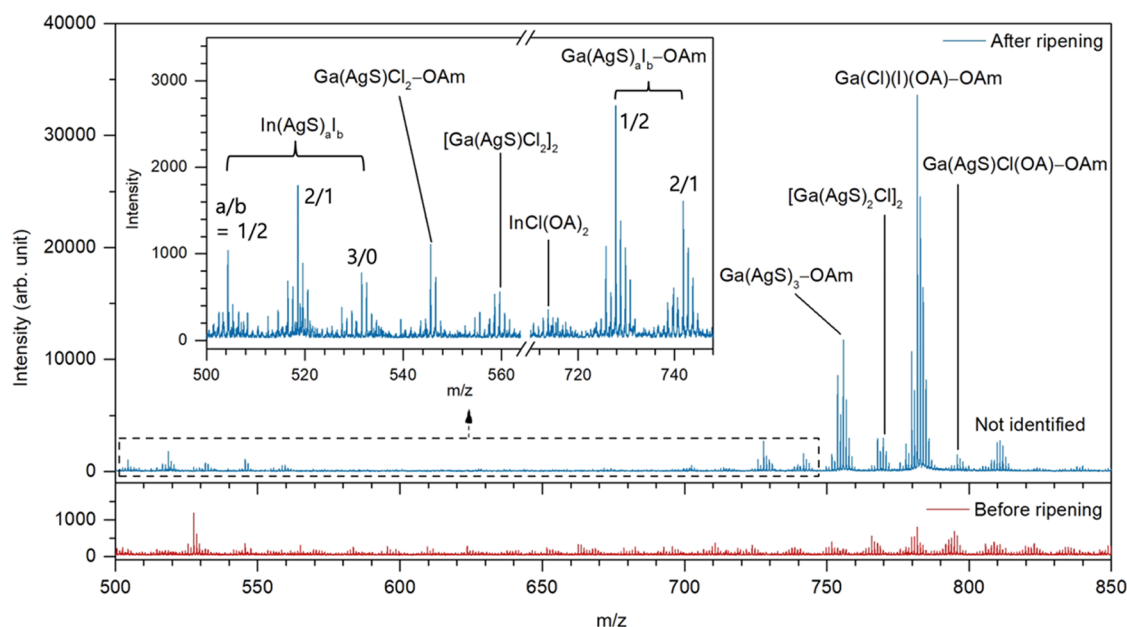
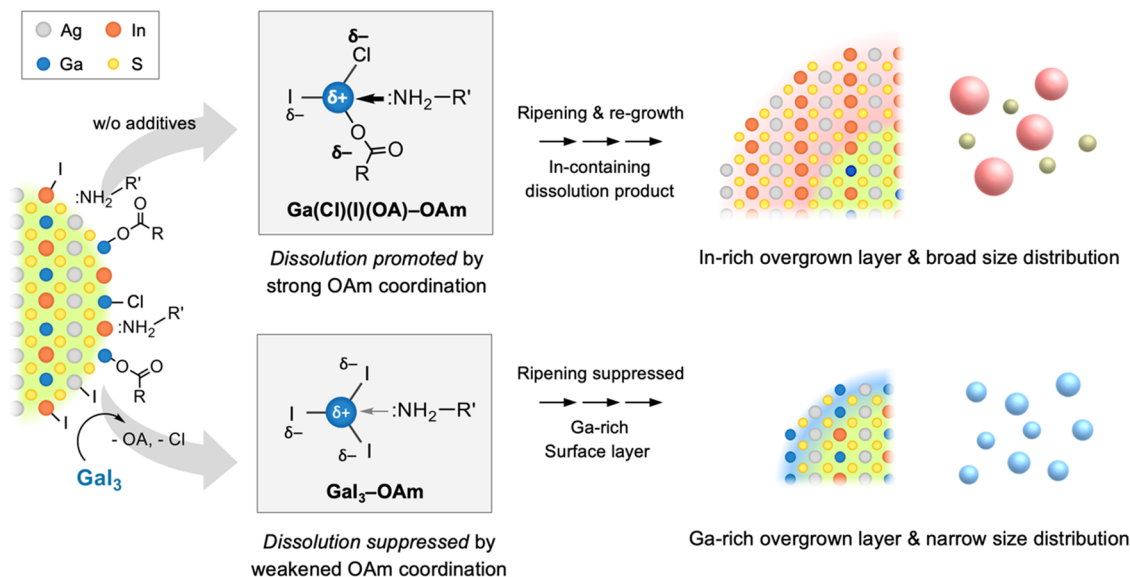


Figure 3. Assignment of dissolution products during Ostwald ripening of AIGS QDs. Mass spectra were acquired using matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) for purified AIGS QDs dispersed in OAm, both before (red traces, lower panel) and after (blue traces, upper panel) ripening at elevated temperature. The inset shows the mass spectra in the m/z ranges of 500–564 and 710–748. Major peaks were assigned to specific molecular species as summarized in Table S3.

Scheme 1. Schematic Illustration of the Halide-Mediated Surface Dissolution and Reincorporation of Dissolved Ga Products during the Ostwald Ripening of AIGS QDs



the dominant products derived from QD dissolution. Second, minor products were also generated in the lower m/z regions of 500–564 and 710–748 (inset of Figure 3). The former includes species such as $\text{In}(\text{AgS})_a\text{I}_b$ or $\text{Ga}(\text{AgS})_a\text{Cl}_b\text{-OAm}$, where halides bind selectively to In^{3+} or Ga^{3+} according to the HSAB theory. The latter includes species such as $\text{Ga}(\text{AgS})_a\text{I}_b\text{-OAm}$, which deviate from the HSAB predictions. In this case, despite the preference of Ga^{3+} for hard anions, the additional coordination of hard OAm appears to stabilize the I^- -containing species. Overall, the majority of identified dissolution products were Ga-containing species coordinated to hard anions ($m/z = 750\text{--}800$), supporting the observed Ga deficiency in the ripened QDs. The dissolved Ga^{3+} species are

strongly bound to hard bases and therefore not readily reincorporated into the QDs during ripening. Instead, they remain in the reaction medium as stable molecular complexes. Notably, although Cl^- originates from InCl_3 that is used as the core precursor, a strong preference for Ga–Cl or more generally Ga–hard anion pairing is observed. This suggests that metal-halide coordination—both on the surface of AIGS QDs and in the medium, either as precursors or as dissolution products—undergoes dynamic exchange, likely governed by HSAB theory. Such a process can even occur during the typical purification of QDs. Although halides were clearly observed in the ripening products, they were not detected by XPS analysis of purified pristine and ripened AIGS QDs (Figure S7 and

Table S1). Since XPS analyzes AIGS QDs after they are separated from the solution, any halide species dissolved in the solution cannot be detected.

Overall, during the ripening process, the constituent atoms dissociate from the surface in the form of atomic clusters associated with OAm. This process leads to the pairing of cations with surface ligands (e.g., OA and halides) and/or AgS units, resulting in their neutralization in the medium. According to the literature, the hardness of an anion modulates the ionicity of its counter metal cation^{34,42} and affects the coordination of cations with surrounding ligands.⁴³ From this point of view, the amplified ripening by the hard Cl^- can be interpreted as the increased ionicity of group III cations in the dissolution product and the corresponding enhanced coordination of OAm. In contrast, the soft I^- lowers the coordination strength of OAm to the dissolution product; thus, QDs can be stabilized after marginal ripening, at which point the surface solubility is balanced with the concentration of the dissolution product. Notably, the accelerated ripening of QDs by Cl^- has also been reported for II–IV⁴⁴ and I–II–VI₂ QDs.¹² A graphical summary of the additive-modified ripening process is presented in Scheme 1.

Crystallographic Disorder Induced by Ostwald Ripening. I–III–VI₂ group semiconductors typically exist in either the tetragonal or orthorhombic crystal structure. The former is known to be thermodynamically stable, while the latter is considered a metastable high-temperature phase.^{45,46} In previous studies on AgInS_2 (AIS) QDs, the orthorhombic phase was observed under high temperatures and large particle sizes.^{47,48} The ripened AIGS QDs experienced rapid size increments at high temperatures, which requires investigation into the stability of crystal structures under given conditions.

We performed X-ray diffraction (XRD) analysis of the ripened QDs with no additives, InCl_3 , and GaI_3 . Each sample was fractionated into two distinct size populations using size-selective precipitation (Figure 4a, see Methods section), which collected more-ripened large QDs from the precipitate and less-ripened small QDs from the supernatant (Figure 4b). The X-ray diffractograms of AIGS and AIGS/AGS QDs exhibited typical tetragonal crystal structures with an intense peak for the (112) plane, which shifted to higher angles as the compressive AGS shell was overcoated. Meanwhile, all ripened QDs revealed additional XRD patterns of the tetragonal (220), (204), and (312) planes (Figure 4c). The shifts in diffraction peaks for various additives reflect the altered chemical composition; for example, an increase in the Ga content leads to a peak shift to a higher angle. No observation for the distinctive peaks in the 30–40° range further supports the exclusion of phase segregation of monoclinic Ag_2S and Ag metal.

Although the dominant diffraction peaks of the ripened AIGS QDs agreed with the tetragonal structure, an orthorhombic phase also emerged from the ripened QDs (orange arrows in Figure 4c). Ripened QDs with InCl_3 or without additives displayed orthorhombic (120) and (121) planes around the tetragonal (112) plane. The development of orthorhombic phase was also observed even at 240 °C (Figure S8) where AGS shell growth is initiated (Figure 1d). Notably, ripening promoted by InCl_3 even resulted in the appearance of unidentified diffraction peaks at 28, 47 and 53.5°, which are seemingly attributed to the (112), (204), and (312) planes of Ga-rich AIGS (indicated by blue arrows in Figure 4c). We speculate that these peaks may originate from secondary

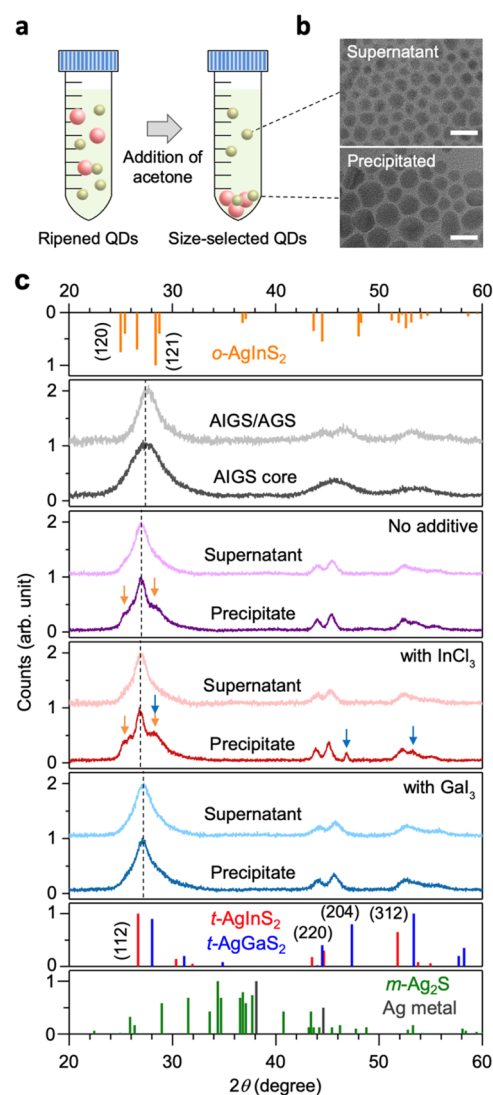


Figure 4. X-ray diffractograms of size-selected AIGS QDs. (a) Illustration of the fractionation of ripened QDs by size-selective precipitation. Centrifugation of QD dispersion with a controlled amount of acetone precipitates the majority of ripened large QDs and leaves less ripened smaller QDs in the supernatant. (b) TEM images of QDs captured from the supernatant (upper) and precipitate (lower). Scale bar: 20 nm. (c) XRD diffractograms of AIGS (top) and AIGS/AGS QDs (bottom) (gray), and ripened QDs in supernatant (top) and precipitate (bottom): without additives (purple), with InCl_3 (red), and with GaI_3 (blue). Orange arrows: orthorhombic (120) and (121) planes. Blue arrows: tetragonal (112), (204), and (312) planes of Ga-rich AgInGaS_2 . The top and bottom panels are reference diffraction patterns for orthorhombic AgInS_2 (JCPDS 00–025–1328), tetragonal AgInS_2 (JCPDS 00–025–1330), AgGaS_2 (JCPDS 00–025–0351), monoclinic Ag_2S (JCPDS 00–014–0072), and Ag metal (JCPDS 00–04–0783).

nucleation of Ga-rich AIGS QDs, derived by the accumulation of Ga-rich dissolution products. Small QDs acquired from the supernatant appeared relatively free from the orthorhombic phase but also exhibited tails relevant to the orthorhombic phase (see the broad lower part of the (112) plane). This observation suggests that the ripening process triggered the formation of an orthorhombic phase owing to an unstable

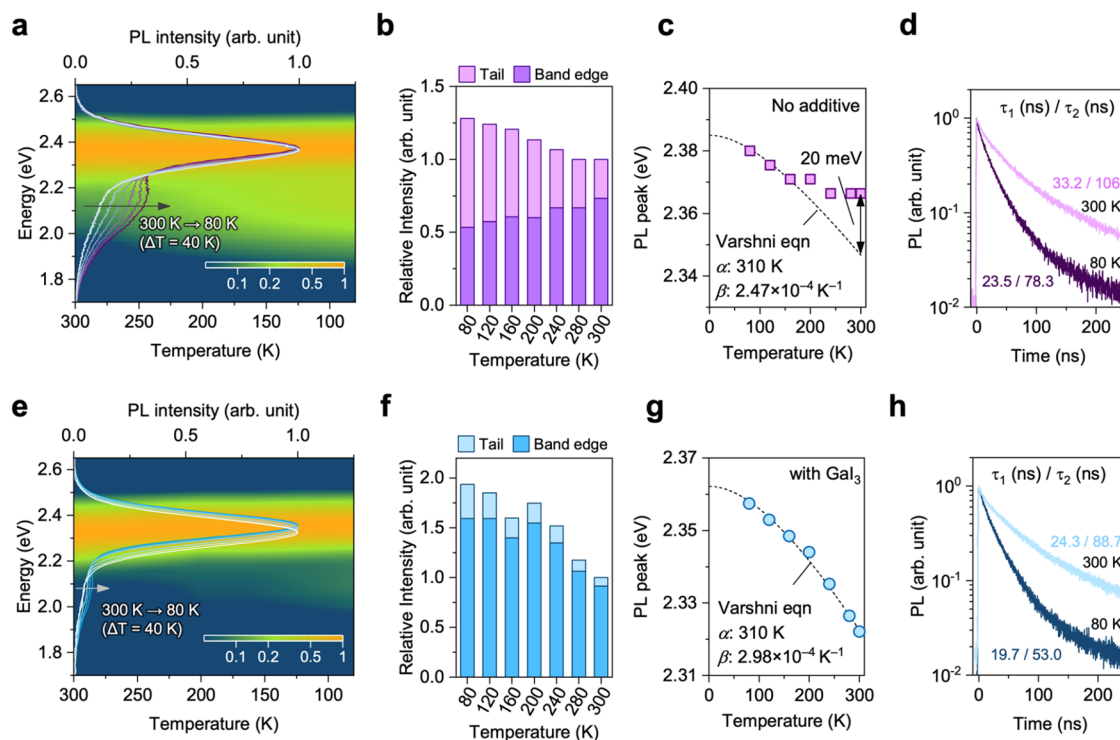


Figure 5. Temperature-dependent PL of AIGS/AGS core/shell QDs. PL properties of AIGS/AGS QDs treated (a–d) without and (e–h) with GaI_3 additive during AGS shell growth. (a, e) PL spectra of AIGS/AGS QDs (a) without and (e) with GaI_3 during AGS shell growth, measured from 300 to 80 K in 40 K intervals. (b, f) Relative intensity of tail (light color) and band-edge emission (dark color) (b) without and (f) with GaI_3 . The data are normalized to unity at 300 K for comparison. (c, g) PL peak energies of AIGS/AGS QDs (c) without and (g) with GaI_3 . Dashed lines indicate the Varshni equation with α and β values indicated in the panel. (d, h) PL decay dynamics recorded at 0.1 eV above the PL peak energy for the QDs (d) without and (h) with GaI_3 at 300 K (light colors) and 80 K (dark colors).

crystal structure, as anticipated from the case of phase transformation in AIS QDs.⁴⁷

This observation raises an additional concern that Ostwald ripening intervened in shell growth and could induce a crystallographic disorder at the core–shell interface, potentially compromising the structural integrity and corresponding optical properties. The batch-to-batch variation in the PL QY and spectral quality observed in AIGS QDs appears to originate from this phase instability, which induces crystallographic defects at the core–shell interface during shell growth. Encouragingly, although this phase instability cannot be eliminated as long as OAm is used, appropriate control of the ripening process can mitigate this issue, for instance, through the addition of GaI_3 , as demonstrated in this study.

Interfacial Defects in AIGS/AGS QDs and Their Influence on PL. In the synthesis of AIGS/AGS core/shell QDs, the excess amount of shell precursors seemingly prevents the ripening of the AIGS cores, as indicated by their size of *ca.* 7 nm. However, the subsisting tail emission in the PL spectra implies the presence of DAP states in the QDs even after AGS shell growth. Our investigation into the ripening onset (Figure 1a–d) and accompanying crystallographic disorder (Figure 4) suggests that the origin of DAP states in AIGS/AGS QDs is surface instability during shell growth, producing interfacial defects. We anticipate the $|D\rangle$ state as Ag_{In} or Ag_{Ga} antisites, and the $|A\rangle$ state as In or Ga vacancies.¹⁷ S vacancies¹⁸ are less likely to be involved in our case since an excess amount of S was provided during the synthesis of QDs. In addition, the consistent Ag-to-S ratio (Figure 1f and Table S2) during ripening supports the exclusion of Ag vacancy as a dominant

origin of DAP emission. The addition of GaI_3 at the beginning of AGS shell growth can stabilize the AIGS surface by delivering iodide ligands and preventing the formation of Ag_{In} or Ag_{Ga} antisites by supplying sufficient Ga during the growth of QDs. In our demonstration, the simple addition of GaI_3 to the shell achieved an improved PL QY of 85% as well as suppressed tail emission, yielding a narrow full width at half-maximum (fwhm) of 30 nm (Figure S9). Nevertheless, a small fraction of residual tail emission ($\sim 5\%$) remains even in the presence of GaI_3 in the shell growth. We attribute this emission to defects remaining within the core interior, possibly originating from the AIGS nanocrystals that were converted from off-stoichiometric Ag_{2-x}S nanocrystals formed in the early stage of the reaction.

Temperature-dependent PL and decay dynamics (Figure 5) provide insights into the defect states involved in the exciton recombination process. The tail emission energy was lowered from the band-edge transition energy, implying that the DAP states are located between the band-edge states. In the comparison of PL spectra at various temperatures, the QDs without GaI_3 (hereafter referred to as “DAP-rich QDs”) exhibited a considerable increase in tail emission with decreasing temperature (Figure 5a), whereas the QDs with GaI_3 (“DAP-less QDs”) barely amplified the tail emission (Figure 5e). This observation suggests that the DAP states are thermally mixed with the band-edge states, despite a considerable energy difference of 250 meV between them. Additionally, even though the DAP states are basically emissive, their generation is related to the formation of nonradiative centers in QDs. The DAP-rich QDs exhibited a

marginal PL increment from 300 to 80 K (approximately 1.25-fold), possibly because of the active involvement of non-radiative pathways that are activated by thermal energy. In contrast, QDs with reduced interfacial disorder by GaI₃ exhibited an approximately 2-fold enhancement in PL intensity over the same temperature range (Figure Sb,f).

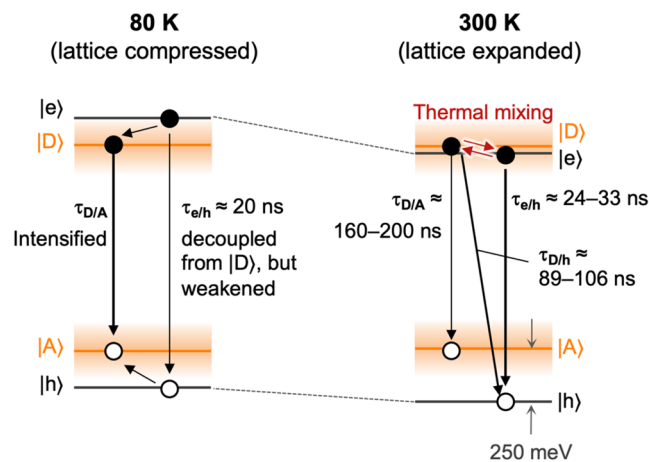
At first glance, the apparent characteristics agree with the generalized picture describing the midgap DAP states in the Al(G)S systems. However, the PL peak shift of the DAP-rich QDs exhibited an abnormal trend that disagrees with this picture. The temperature dependence of E_g is typically interpreted by the Varshni equation⁴⁹ [$E_g(T) = E_g(0) - \alpha T^2/(\beta + T)$, where T is the temperature, α is the Debye temperature, and β is an empirical fitting parameter]. While the DAP-less QDs followed the expected thermal behavior with $\alpha = 310$ K (Figure 5g, deduced from a linear combination of $\beta_{\text{AIS}} = 340$ K and $\beta_{\text{AGS}} = 280$ K with a Ga-to-In ratio of 1.22), the QDs containing interfacial defects displayed abnormal saturation of the PL peak energies as the temperature increased (Figure 5c), *ca.* 20 meV larger than the expected value at 300 K. According to the generalized picture, if the PL peak is tarnished by the DAP states, this value should be smaller.

To gain more insight, we investigated the temperature-dependent PL decay dynamics. By choosing the detection energy using a monochromator, we can discriminate emissions from the DAP states and band-edge states of the QDs. We separated the decay dynamics into two components using a biexponential fit (fast τ_1 and slow τ_2 ; see Figure 5d,h). For the tail emission associated with the DAP states (Figure S10), we observed an ultrafast pathway seemingly related to non-radiative centers (τ_1 , not reliably resolved due to limited resolution) and a characteristic DAP recombination pathway with $\tau_2 \approx 200$ ns. τ_2 showed a weak temperature dependence, increasing the decay rate upon increased temperature, due to the highly localized nature of carriers in DAP states.^{50,51} Meanwhile, for the emission at +0.1 eV from the PL peak (Figure 5d,h), which minimized the involvement of DAP emission, both types of QDs displayed accelerated exciton recombination upon lowering the temperature, but the DAP-less QDs showed faster decay rates ($\tau_1/\tau_2 = 23.5$ ns/78.3 ns for DAP-rich QDs versus 19.7/53.0 ns for DAP-less QDs at 80 K). Coupled with the temperature-dependent PL peak energies, this DAP-dependent recombination dynamics suggests the thermal mixing of band-edge states with DAP states at 300 K, which places the higher |D> than the band-edge electron state (|e>) or the lower |A> than the band-edge hole state (|h>). Such an electronic structure captures band-edge carriers to trap states in assistance with thermal energy and retards exciton recombination due to either slow detrapping^{52,53} or reduced wave function overlap of localized charges.⁵⁴ Also, the |D>–|h> or |A>–|e> recombination leads to a PL peak energy increase of approximately 20 meV at 300 K (Figure 5c), which is comparable to the thermal energy at this temperature (26 meV). This phenomenon is relaxed at lower temperatures, where carriers at the band-edge states are seldom excited to the DAP states.

According to a theoretical study,⁵⁵ the negative deformation potential of tetragonal phases (*e.g.*, $dE_g/dT = -2.2$ for AGS) predicts an elevation of conduction and valence bands as temperature decreases due to lattice compression. In contrast, the temperature-dependent shift of |D> or |A> states would be diverse, for example, the dopant states in GaN.⁵⁶ Although an in-depth study on the temperature dependence of |D> and |A>

in AIGS is lacking, in Scheme 2 we propose a hypothetical energy level diagram that places |D> above |e> to explain the

Scheme 2. Proposed Energy Level Diagram of AIGS/AGS QDs Containing Donor (|D>) and Acceptor (|A>) States at 80 K (Left) and 300 K (Right)



enhanced tail emission (Figure 5a,b) and reduced thermal mixing of |D> above |e> (Figure 5c) at lower temperatures. At $T = 300$ K, the higher |D> accepts photogenerated electrons from |e> by thermal excitation, and their relative occupation leads to two decay pathways observed in Figure 5d, in which the fast τ_1 corresponds to the |e>–|h> transition ($\tau_{e/h} \approx 24$ –33 ns, influenced by the amount of DAP states) and slow τ_2 to the |D>–|h> transition ($\tau_{D/h} \approx 89$ –106 ns). Because of the higher thermal population of the lower |e> state and faster |e>–|h> transition, the static PL spectra exhibit a marginal contribution of tail emission associated with the |D>–|A> transition. Meanwhile, at a reduced temperature of 80 K, |e> is raised above |D>, and the extent of this elevation is at least comparable to or greater than the PL blueshift of |e>–|h> transition (*ca.* 40 meV, as anticipated from Figure 5g), since the blueshift results from the upward shift of both |e> and |h> levels.⁵⁵ Considering that the energy difference between |D> and |e> is *ca.* 20 meV (Figure 5c), this elevation is sufficient to decouple |e> from |D> at 80 K with a thermal energy of 6.9 meV. This altered electronic structure increases the occupation of |D> and |A> states by photogenerated carriers to and consequently amplifies tail emission. Even though the |e>–|h> transition is weakened, the decoupling between |D> and |e> recovers the intrinsic recombination rate of the |e>–|h> transition, as inferred from the convergence of fast τ_1 to *ca.* 20 ns in Figure 5d,h. Notably, the alternative scenario in which |A> lies below |h> fails to explain the temperature-dependent behavior, as it would predict suppressed tail emission at 80 K, contrary to our observations.

CONCLUSIONS

In this study, we uncovered the critical impact of surface instability in AIGS QDs on the structural and optical integrity of AIGS/AGS core/shell heterostructures. We demonstrate that Ostwald ripening, promoted by the strong coordination affinity of OAm, induces a Ga-deficient interface and crystallographic disorder, leading to the formation of interfacial DAP states and consequent tail emission. By leveraging the coordination strength of OAm to the potential dissolution

products using GaI_3 as an additive during shell growth, we successfully suppressed the undesired Ga extraction and crystallographic disorder in the heteroepitaxy of AGS shells on AIGS cores. This surface reinforcement by GaI_3 treatment led to a PL QY of 85%, suppressed tail emission, and a narrow emission line width of 30 nm. Temperature-dependent PL analysis allowed us to propose a hypothetical electronic structure of the AIGS/AGS QDs containing DAP states. Specifically, the uncontrolled surface instability during shell growth leaves the thermally mixed $|D\rangle$ above the $|e\rangle$ state, which makes tail emission induced by $|D\rangle$ – $|A\rangle$ transition less visible at 300 K. At lower temperatures, the amplified tail emission and recovered $|e\rangle$ – $|h\rangle$ transition can be attributed to the elevation of $|e\rangle$ and $|h\rangle$ states, which in turn is explained by the negative deformation potential of the tetragonal structure.

Although the main culprit of tail emission locates at the interface and the GaI_3 effectively mitigates its emergence, the small degree of tail emission still remains in the PL spectra of AIGS/AGS QDs (Figure S9). We attribute this residual emission to defects within the interior of QDs, rather than at the interface. Although the proposed synthetic pathway involves the formation of AIGS cores through an Ag – S – Ga –(oleate)₂ molecular complex and an In precursor, a competing pathway involving the formation of Ag_2S followed by cation exchange with In and Ga ¹³ is also plausible, due to the chemical instability of Ag that readily forms Ag_2S in the early stages of the reaction. Previous studies have shown that the synthesis of Ag_2S nanocrystals and nanostructures frequently results in off-stoichiometric Ag_{2-x}S .⁵⁷ The cation exchange of such nanocrystals with In and Ga may leave behind cation vacancies that acts as $|A\rangle$. To suppress the formation of such defect, we believe that following strategies would be beneficial: (i) careful control of early reaction stage to prevent the formation of off-stoichiometric Ag_{2-x}S ; (ii) guiding the reaction exclusively through I–III–VI₂ type intermediates such as Ag – S – Ga –(oleate)₂. In addition to these fundamental strategies, increasing the reactivity of group III precursors may also be helpful [e.g., gallium tris(*N,N'*-diethyldithiocarbamate)],¹³ as it would allow sufficient incorporation of group III cations before the Ag_{2-x}S domains grows too large.

This work was originally intended to pursue trap-free, high-performance AIGS QDs by improving the crystallographic and compositional integrity of core/shell heterostructures. Beyond offering a new perspective on the synthesis of heterostructured AIGS/AGS QDs, we believe that our findings also provide general mechanistic insight into the synthesis of various nanocrystals carried out in coordinating media. Given the widespread use of alkylamine-based syntheses employing metal halides across II–VI, III–V, and I–III–VI₂ QDs, the principles identified here are broadly applicable—particularly in reactions involving strong coordination between metal halides and participating ligands. Such insights may prove important in guiding the synthesis of nanocrystals with previously unexplored chemical compositions.

METHODS

Materials. Silver(I) iodide (AgI , 99.9%), gallium(III) iodide (GaI_3 , 99.999%), indium(III) chloride (InCl_3 , 99.995%), indium(III) iodide (InI_3 , 99.995%), and gallium(III) chloride (GaCl_3 , 99.99%) were purchased from Thermo Fisher Scientific. Sulfur powder (S_8 , 99.9%), oleic acid (OA, 99%), and *n*-triethylphosphine (TOP, 99%) were purchased from Uniam. Tetrabutylammonium iodide (TBAI,

98%), tetrabutylammonium chloride (TBACl, 98%), gallium(III) acetylacetonate ($\text{Ga}(\text{acac})_3$, 99.99%), oleylamine (OAm, 98%), 1-dodecanethiol (DDT, 98%), tetrahydrofuran (THF, 99.8%) and all solvents including toluene (anhydrous 99.8%), ethanol (anhydrous, 99.5%), and chloroform (99.8%) were purchased from Sigma-Aldrich.

Preparation of Precursors. A 0.5 M gallium oleate ($\text{Ga}(\text{OA})_3$) solution was prepared by degassing a mixture of 10 mmol of $\text{Ga}(\text{acac})_3$, 10 mL of OA, and 10 mL of ODE at 130 °C for 4 h. A 0.2 M AgI solution was prepared by dissolving AgI (2 mmol) in 10 mL of OAm. A 0.3 M InCl_3 solution was prepared by dissolving 3 mmol of InCl_3 in 10 mL of THF at 50 °C. A 1.0 M sulfur in OAm (S-OAm) was prepared by reacting sulfur powder (10 mmol) with OAm (10 mL) at room temperature. All the reactions were performed under an inert atmosphere.

Synthesis of AIGS Cores. The synthesis protocol of AIGS cores followed the literature method^{11,20} with a minor modification. All syntheses were carried out under an inert atmosphere using a Schlenk line. InCl_3 in THF (0.3 M, 0.33 mL), AgI in OAm (0.2 M, 0.5 mL), and OAm (5 mL) were placed in a 100 mL three-neck flask and evacuated at 50 °C for 1 h. After backfilling with N_2 , S-OAm (1.0 M, 0.8 mL) was rapidly injected into the flask, followed by the addition of $\text{Ga}(\text{OA})_3$ (0.5 M, 0.3 mL) and DDT (0.25 mL). The reaction mixture was then heated to 210 °C and held for 30 min to grow the AIGS cores.

AGS Shell Growth on AIGS Cores. The as-synthesized AIGS cores that had not undergone intentional ripening were used without any purification steps. To grow the AIGS shell on the cores, AgI in OAm (0.2 M, 0.25 mL), $\text{Ga}(\text{OA})_3$ (0.5 M, 2 mL), and S-OAm (1.0 M, 2.5 mL) were added to the crude core solution at 210 °C. For GaI_3 -mediated surface reinforcement of AIGS cores, GaI_3 solution (0.05 M, 1 mL) was also added together with the shell precursors. After raising the temperature to 280 °C over 10 min, GaCl_3 dissolved in OAm (0.5 M, 1.0 mL) was injected, and the reaction was maintained for 30 min to complete shell growth. All synthetic procedures were performed under an inert atmosphere.

Purification of QDs. To prevent oxidation, the purification of QDs was performed inside a N_2 -filled glovebox. The purification was followed by a precipitation/redispersion procedure. Specifically, 10 mL of toluene and 30 mL of ethanol were added to 5 mL of the crude solution in a conical tube to flocculate QDs, and the mixture was centrifuged at 7878 g for 10 min. After discarding clear supernatant, the precipitate was redispersed in 10 mL of toluene. The second precipitation was performed by adding 30 mL of ethanol, followed by centrifugation at 7878 g for 10 min. The final precipitate was redispersed in hexane for further analysis.

Intentional Ripening of AIGS Cores with Metal Halides.

Intentional ripening was performed directly in the reaction vessel containing the as-synthesized AIGS cores. After quenching the AIGS core solution to room temperature, a metal halide (GaCl_3 , GaI_3 , InCl_3 , or InI_3) dissolved in OAm (0.05 M, 1 mL) was injected into the reactor, followed by reaction at 210 °C for 10 min. To initiate the ripening process, the mixture was heated to 280 °C and maintained for 30 min. Aliquots were periodically collected from the reaction mixture to measure the absorption and PL spectra.

Size-Selective Fractionation of Ripened AIGS QDs for XRD Analysis.

Size-selective fractionation and subsequent purification were performed in a glovebox. For fractionation, 8 mL of the crude QD solution was transferred to a conical tube, and 30 mL of acetone was added. The mixture was centrifuged at 7878g for 10 min to precipitate the larger QDs. The precipitation was redispersed in toluene (10 mL) and further purified by adding ethanol (30 mL), followed by centrifugation at 7878g for 10 min. The supernatant containing small QDs was carefully transferred to conical tubes. To isolate the smaller QDs, 30 mL of ethanol was added to each 10 mL portion of the supernatant, followed by centrifugation at 7878g for 10 min. The resulting precipitate was redispersed in 10 mL of toluene and subjected to an additional purification step using ethanol precipitation and centrifugation, yielding purified smaller-sized QDs.

Optical Spectroscopy. Absorption spectra of the QDs were measured using a Shimadzu UV-2600i ultraviolet–visible spectrom-

eter. Static PL spectra and PL QY were investigated using a Horiba FluoroMax Plus-C spectrofluorometer (spectral accuracy: ± 0.5 nm) equipped with a 3.2-in. integrating sphere (excitation: 450 nm; PL QY accuracy: $\pm 3\%$).

X-ray Photoelectron Spectroscopy. Ga 3d and In 4d photoelectron spectra of pristine and ripened AIGS cores and AIGS/AGS QDs were obtained using a Thermo Fisher Scientific NEXSA instrument. Thin QD films were fabricated on a Si substrate by spin-coating a purified QD dispersion. The samples were prepared under an inert atmosphere to avoid oxidation of the QDs.

Morphology Analysis. The size and morphology of the QDs were estimated using a JEOL JEM-3010 transmission electron microscope (TEM). The average diameter and standard deviation were determined from the geometric average of more than 50 individual QDs, by converting their projected area into an equivalent circular diameter (d) using the formula $d = 2 \times (\text{area}/\pi)^{1/2}$.

Crystallography Analysis. XRD patterns were acquired with a Rigaku SmartLab diffractometer using a Cu K α source ($\lambda = 1.5418$ Å). Size-fractionated QDs were dried overnight under vacuum to obtain powdered samples. The powders were mounted on a glass holder for XRD measurements.

Chemical Composition of QDs. The cation composition in the AIGS QDs was estimated using inductively coupled plasma-atomic emission spectroscopy (ICP-AES, PerkinElmer OPTIMA 8300). The QD powder was digested in nitric acid prior to analysis. To prevent the potential loss of sulfur species during ICP-AES, the overall elemental composition was also assessed using energy-dispersive X-ray spectroscopy (EDS, JEOL JEM-F200).

Mass Spectrometry. Mass spectra of the dissolution products were acquired using a matrix-assisted laser desorption/ionization-time-of-flight (MALDI-TOF) mass spectrometer (Applied Biosystems Voyager DE-STR). The purified AIGS QDs were redispersed in 7 mL of OAm, the resulting dispersion was evacuated at 100 °C for 1 h. The QDs were then ripened at 280 °C for 30 min and cooled down to room temperature. A dithranol matrix was mixed with the QD solution, both before and after the ripening, diluted in chloroform, and the resulting mixture was deposited onto the MALDI target. Ionization was performed using a pulsed nitrogen laser (337 nm, 3 ns pulse width). An accelerating voltage of 20 kV was applied to detect the resulting ions.

Temperature-Dependent PL. Temperature-dependent PL spectra and decay dynamics were measured using a Horiba FluoroMax Plus-C spectrofluorometer equipped with a home-built liquid N₂ cryostat and a Delta Time time-correlated single-photon counting module. A pulsed laser diode (excitation wavelength: 402 nm; pulse width: ~ 1 ns; repetition rate: 1 MHz) was used as the excitation source. For low-temperature analysis, QD dispersions in hexane were spin-coated onto cover glasses and mounted on the cryostat holder. The sample was cooled to 80 K using liquid N₂ and further stabilized for 30 min. The PL spectra and decay dynamics were acquired as the temperature was increased stepwise from 80 to 120, 160, 200, 240, 280, and 300 K.

ASSOCIATED CONTENT

Data Availability Statement

The data supporting the findings of this study are available from the corresponding authors upon reasonable request.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.5c07418>.

Absorption, PL and TEM characterizations of QDs; temperature-dependent PL QY evolution; EDS element maps and atomic compositions; XPS spectra of surface halides; XRD pattern of AIGS QDs ripened at 240 °C; temperature-dependent PL decay dynamics probed at the trap emission; and peak assignments in MALDI-TOF mass spectra (PDF)

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Author Contributions

S.P., B.J.K., W.H.J., K.H.L., and H.L. conducted the synthesis and characterization of the materials. S.P. and B.J.K. performed the spectroscopic analyses. S.P. and H.L. drafted the manuscript. H.L. and J.L. conceived the study and supervised the experiments, data analysis, and manuscript preparation. All authors contributed to manuscript preparation and reviewed the final version.

Notes

The authors declare no competing financial interest.

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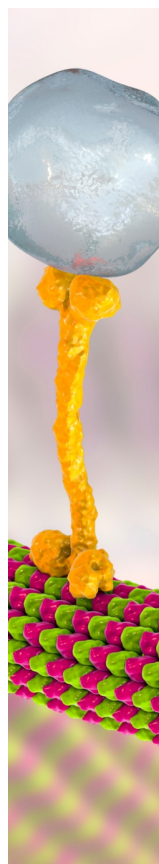
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